Why We Backtest

What a sample can tell you, and what it cannot

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An experiment

Take 49 industry portfolios. Every December:

1. Regress each industry on the market, using the last three years of daily returns

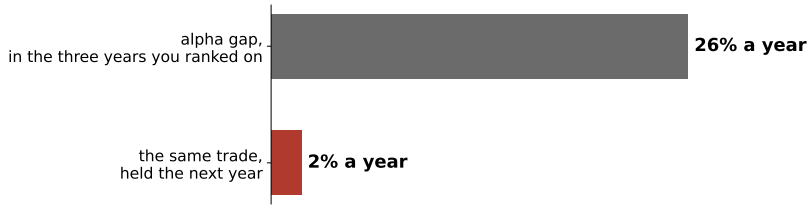
$$r_t = \alpha + \beta\, r_{m,t} + \varepsilon_t$$

2. Buy the fifth with the highest α, short the fifth with the lowest
3. Hold for the next year

Would that work?

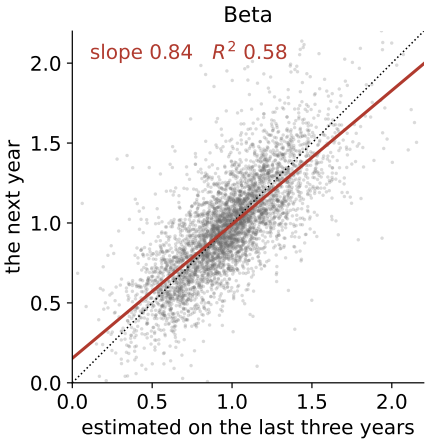
Details

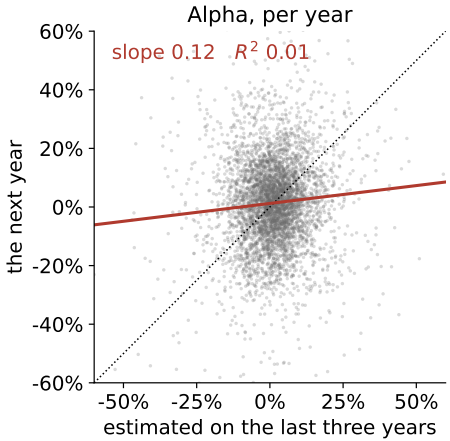
Promised 26%, delivered 2%



- More than nine-tenths of the alpha you ranked on never showed up
- And 2% a year is not distinguishable from zero: $t = 1.3$ over 97 years
- The trade is hedged: appraisal ratio = average return / volatility $= 1.9\% / 14.3\% = 0.13$
- The same number from the t and the sample size alone: $t/\sqrt{T} = 1.3/\sqrt{97} = 0.13$

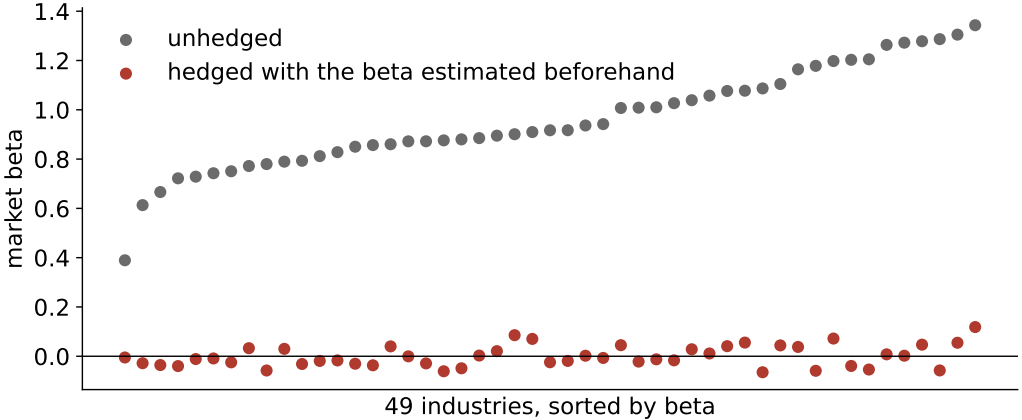
The other coefficient behaves





- Same regression, same three years. Beta carries over; alpha does not.

So a hedge built in advance works



- Hedge set before the year starts: betas of 0.39 to 1.34 fall to 0.03 on average

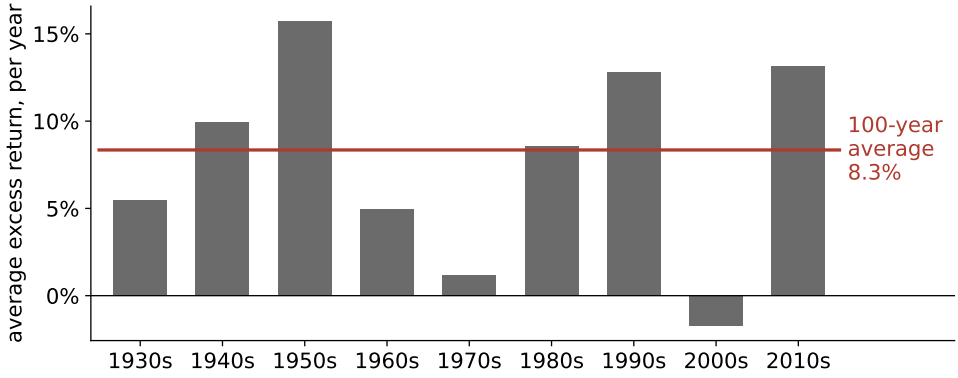
Three years measure beta, not alpha

49 industries, three years of daily data each, 1927–2025:

	average	standard error	tested against	share with $\lvert t \rvert > 2$
β	0.97	0.03	the average β	82%
α	1.4% a year	6.9%	zero	12%
$\alpha/\sigma_\varepsilon$ (appraisal ratio)	0.10	0.58	zero	12%

- Four betas in five are measurably different from the average beta
- The alpha's standard error is five times the alpha
- The appraisal ratio's standard error is $1/\sqrt{3} = 0.58$, whatever the industry

How long to learn a mean: the market



- 100 years: mean 8.3%, volatility 18.4%, Sharpe ratio 0.45
- After *ten* years, the standard error of the mean is still 5.8%

The arithmetic

Each year's return has variance σ^2, and years are independent. The mean over T years:

$$\text{Var}\Big(\frac{1}{T}\sum_{t=1}^{T} r_t\Big) \;=\; \frac{1}{T^2}\times T\,\sigma^2 \;=\; \frac{\sigma^2}{T} \qquad\Longrightarrow\qquad \text{standard error} \;=\; \frac{\sigma}{\sqrt{T}}$$

$$t \;=\; \frac{\text{mean}}{\sigma/\sqrt{T}} \;=\; \text{ratio}\times\sqrt{T} \qquad\Longrightarrow\qquad T \;=\; \Big(\frac{z}{\text{ratio}}\Big)^2$$

- **ratio**: the Sharpe ratio for a mean, the appraisal ratio for an alpha (annual)
- z: the t-statistic you require; T: years of data

The market: $(1.64/0.45)^2 = $ **13 years** to be 95% sure its premium is above zero

Years of data to reject zero

	t-statistic required			
Sharpe or appraisal ratio	1.64	1.96	2.58	3.00
0.30	30	43	74	100
0.45 (the market)	13	19	33	44
0.50	11	15	27	36
0.75	4.8	6.8	12	16
1.00	2.7	3.8	6.7	9.0
1.50	1.2	1.7	3.0	4.0
2.00	0.7	1.0	1.7	2.3
3.00	0.3	0.4	0.7	1.0

- Halve the ratio and you need four times the data

The same arithmetic, with a hurdle

Divide by σ: the ratio itself has standard error $1/\sqrt{T}$. To show it is above a hurdle, not just above zero:

$$t = \frac{\text{ratio} - \text{hurdle}}{1/\sqrt{T}} = (\text{ratio} - \text{hurdle}) \times \sqrt{T} \qquad \Longrightarrow \qquad T = \left(\frac{z}{\text{ratio} - \text{hurdle}}\right)^2$$

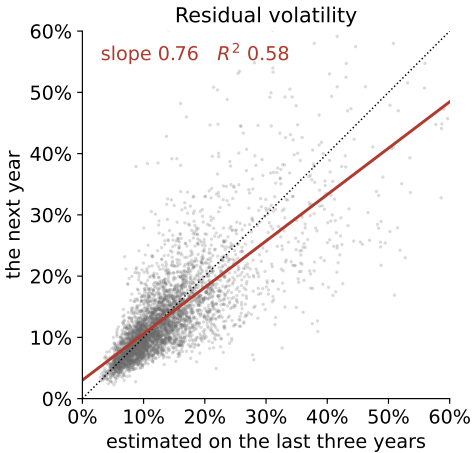
- Only the gap between the ratio and the hurdle counts
- A hurdle of zero gives back the formula you just used
- A ratio of 1 against a hurdle of 0.5: $(1.64/0.5)^2 = 11$ years

Years to clear a hurdle

	to be 95% sure the ratio is above			
True ratio	0	0.25	0.50	1.00
0.30	30	1,076	—	—
0.50	11	43	—	—
0.75	4.8	11	43	—
1.00	2.7	4.8	11	—
1.50	1.2	1.7	2.7	11
2.00	0.7	0.9	1.2	2.7
3.00	0.3	0.4	0.4	0.7

- The closer the truth is to the bar, the longer it takes

Beta and volatility are learned fast



- Betas and volatilities are about how returns *move*, not where they end up
- Their precision grows with the number of observations: sample daily, and months are enough
- A mean only gets sharper with calendar time

Frequency helps beta and volatility. The mean needs time.

T years of data, cut into N observations. Daily instead of monthly: same T, 21 times the N.

	standard error	shrinks with
mean	$\sigma / \sqrt{T}$	years
alpha	$\sigma_\varepsilon / \sqrt{T}$	years
beta	$(\sigma_\varepsilon / \sigma_f) / \sqrt{N}$	observations
volatility	$\approx \sigma / \sqrt{2N}$	observations

- Three years of industry returns, monthly to daily: beta's standard error falls from 0.14 to 0.03. Alpha's barely moves: 7.3% to 6.9%.
- The average return uses only the first price and the last. Looking in between adds nothing.

Back to the factor model

$$r_t = \alpha + \beta f_t + \varepsilon_t$$

	what it is	learned in	do recent past predicts future?
β	exposure to the factor	months	yes
σ_ε	risk of its own	months	yes
$E[f]$	factor premium	decades	no
$\sigma[f]$	factor vol	months	yes
α	the edge	decades?	no

- The risk side of the model is knowable. The return side is the hard part.

And alpha is fleeting

- An edge that other people can see gets traded away
- So an alpha is a moving target: by the time the sample is long enough to measure it, it may be gone
- One more experiment

Ideas cannot come from the data alone

You need **decades** of data to measure an alpha that may last only **years**.

- So the data cannot be where the idea comes from
- The reason comes first: who is on the other side, and why are they paying you?
- Then the data tests it

Estimate, tune, test

sample	what you do with it	may you look?
Estimate	fit: the alphas, the betas, the ranking	yes
Tune	choose among your rules	yes
Test	report	once, at the end

One question: are my estimates stable enough to trade on data they have not seen?

- The experiment was exactly this: estimate on three years, test on the next

How long should each window be?

window	long enough that	so check
Estimate	the parameters are measured well	their standard errors
Tune	the metric you choose on is measured well	or you are choosing on noise
Test	the verdict is not noise	the standard error of the result

- In every window, a ratio measured over T years has a standard error of about $1/\sqrt{T}$

An example: looking for an appraisal ratio of 1

You want to be 95% sure the appraisal ratio is above zero.

$$T = \left(\frac{1.64}{\text{ratio} - \text{hurdle}}\right)^2 = \left(\frac{1.64}{1-0}\right)^2 = 2.7 \text{ years}$$

2.7 years in *each* window: about **8 years** of data in all

Years of data you need, all three windows

	to be 95% sure the ratio is above			
True appraisal ratio	0	0.25	0.50	1.00
0.50	32	129	—	—
0.75	14	32	129	—
1.00	8.1	14	32	—
1.50	3.6	5.2	8.1	32
2.00	2.0	2.6	3.6	8.1
3.00	0.9	1.1	1.3	2.0

- The notebook has 21 years, split 10 / 5 / 6: standard errors of 0.32, 0.45 and 0.41

Something to think about

The same experiment, with one year of data

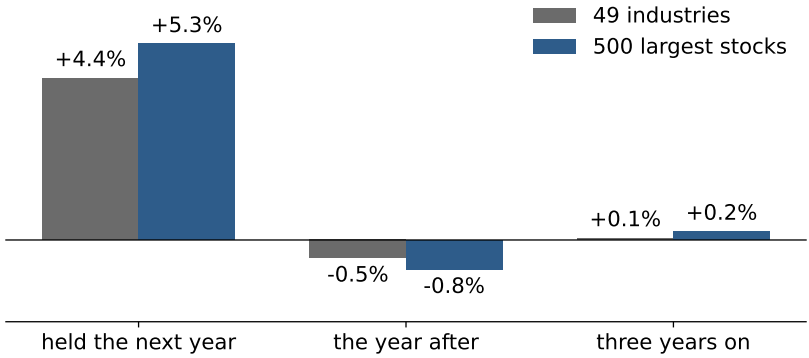
Sorted into thirds here: buy the third with the highest α, short the third with the lowest.

estimated on	alpha's standard error	promised	delivered next year	t
the last three years	6.9%	20%	1.7%	1.6
the last year only	11.6%	33%	4.4%	4.2

- A third of the data, and a noisier alpha
- Yet the trade earns more than twice as much

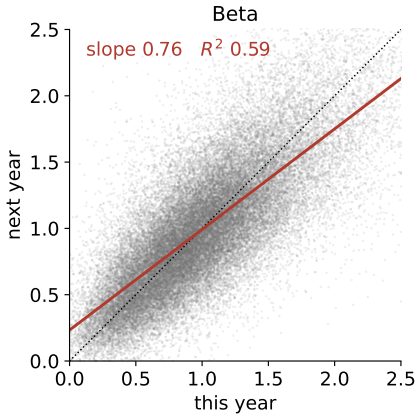
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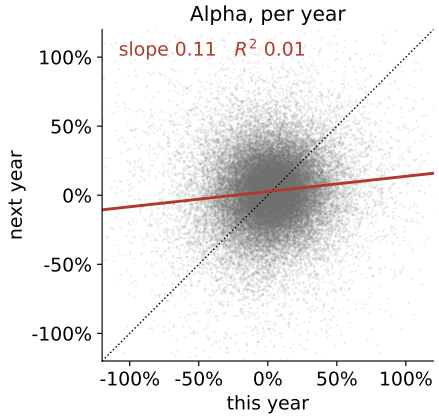
Rank on one year, then wait



- Ranked on a single year: 4–5% the next year, and nothing after that

The same on the 500 largest stocks





- One year of data per stock: promised 53%, delivered 5% the next year

Details

What is going on?

How can a **smaller** sample lead to a **better** result?

- Is alpha better measured with one year of data than with three?
- If not, what is going on?

Where this lives in the notebook

the idea	section of *Backtesting: Estimate, Tune, Test*
In-sample winners disappoint	Live demo: estimate $\rightarrow$ tune $\rightarrow$ test
A track record you could have lived through	2. Walk-forward
How precisely a Sharpe ratio is measured	3. What you report at the end
More attempts, higher bar	5. You tried more than one
How long before you act	7. When to start, and when to stop

- The code in each section computes one of these. Know which before you read it.

Appendix

A1: How the experiment is built

- **Data**: Ken French's 49 industry portfolios, daily returns, 1927–2025
- **Each December**: one regression per industry on the market, over the last three calendar years
- **The trade**: top fifth by estimated α minus bottom fifth, equal-weighted, held for the next year
- **The hedge**: each industry minus its estimated β times the market
- No trading costs

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A2: How the stock version is built

- **Universe**: each year, the 500 largest US common stocks by market value at the end of the previous year (CRSP)
- **Regression**: daily returns within one calendar year, 1927–2025, delisting returns included
- **Same trade**: top third by α minus bottom third, hedged with the ranking year's β
- Each year's picks are followed for three more years, whether or not they stay in the top 500

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